Numerical Linear Algebra
Lecture 3: Orthogonality and Vector Norms

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Outline

1 Inner product and orthogonal vectors

2 Unitary matrices

3 Vector Norms

Inner product and orthogonal vectors

Transpose and Adjoint

- Transpose of $\mathbf{A}$, denoted by $\mathbf{A}^{\mathbf{T}}$, is the matrix $\mathbf{B}$ with $b_{ij} = a_{ji}$
- *Adjoint* or *Hermitian conjugate*, denoted by $\mathbf{A}^*$ or $\mathbf{A}^H$, is the matrix $\mathbf{B}$ with $b_{ij} = \bar{a}_{ji}$
- Note that, $(\mathbf{AB})^T = \mathbf{B}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}}$ and $(\mathbf{AB})^* = \mathbf{B}^*\mathbf{A}^*$
- A matrix $\mathbf{A}$ is symmetric if $\mathbf{A} = \mathbf{A}^{\mathbf{T}}$ (i.e., $a_{ij} = a_{ji}$). It is *Hermitian* if $\mathbf{A} = \mathbf{A}^{\mathbf{T}}$ (i.e., $a_{ij} = \bar{a}_{ji}$)
- For $\mathbf{A} \in \mathbb{R}^{m\times n}$, null($\mathbf{A}$) and range($\mathbf{A}^{\mathbf{T}}$) are complementary subspaces. In addition, null($\mathbf{A}$) and range($\mathbf{A}^{\mathbf{T}}$) are orthogonal to each other (to be explained later)
- For $\mathbf{A} \in \mathbb{C}^{m\times n}$, null($\mathbf{A}$) and range($\mathbf{A}^*$) are complementary subspaces

Inner Product

- Inner product (dot product) of two column vectors $\mathbf{u}$, $\mathbf{v} \in \mathbb{C}$ is $\mathbf{u}^*\mathbf{v}$
- In contrast, *outer* product of $\mathbf{u}$ and $\mathbf{v}$ is $\mathbf{u}\mathbf{v}^*$
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Different ways to see the inner product

1. Vector-vector multiplication: $\mathbf{u}^*\mathbf{v} = \sum_{i=1}^{m} \bar{u}_i v_i$
2. Euclidean length of $\mathbf{u}$ is the square root of the inner product of $\mathbf{u}$ with itself, i.e., $\sqrt{\mathbf{u}^*\mathbf{u}}$
3. Inner product of two unit vectors $\mathbf{u}$ and $\mathbf{v}$ is the cosine of the angle α between $\mathbf{u}$ and $\mathbf{v}$, i.e., $\cos\alpha = \frac{u^* v}{||u||||v||}$

Inner Product is bilinear

Inner product is *bilinear*, in the sense that it is linear in each vertex separately:

- $(u_1 + u_2)^* v = u_1^* v + u_2^* v$
- $u^*(v_1 + v_2) = u^* v_1 + u^* v_2$
- $(\alpha u)^*(\beta v) = \bar{\alpha}\beta u^* v$

Orthogonal Vectors

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In other words, the angle between them is 90 degrees

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Definition

A set of nonzero vectors S is *orthogonal* if they are pairwise orthogonal. They are *orthonormal* if it is orthogonal and in addition each vector has unit Euclidean length.

Orthogonal Vectors

Theorem

The vectors in an orthogonal set S are linearly independent.

Proof

Prove by contradiction. If a vector can be expressed as linear combination of the other vectors in the set, then it is orthogonal to itself

Question: If the column vectors of an $m \times n$ matrix $\mathbf{A}$ are orthogonal, what is the rank of $\mathbf{A}$?

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Question: If the column vectors of an $m \times n$ matrix **A** are orthogonal, what is the rank of **A**?
Answer: $n = min\{m, n\}$. In other words, **A** has full rank

Components of Vector

- Given an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_m\}$ forming a basis of $\mathbb{C}^m$, vector $\mathbf{v}$ can be decomposed into orthogonal components as $\mathbf{v} = \sum_{i=1}^{m} (\mathbf{q}_i^* \mathbf{v}) \mathbf{q}_i$

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- Another way to express the condition is $\mathbf{v} = \sum_{i=1}^{m} (\mathbf{q}_i^* \mathbf{q}_i) \mathbf{v}$
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- $\mathbf{q}_i \mathbf{q}_i^*$ is an *orthogonal projection matrix*. Note that it is NOT an orthogonal matrix.
- More generally, given an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_n\}$ with $n \leq m$, we have

$$\mathbf{v} = \mathbf{r} + \sum_{i=1}^{n} (\mathbf{q}_i^* \mathbf{v}) \mathbf{q}_i = \mathbf{r} + \sum_{i=1}^{n} (\mathbf{q}_i \mathbf{q}_i^*) \mathbf{v} \text{ and } \mathbf{r}^* \mathbf{q}_i = 0, \ 1 \leq i \leq n$$

- Let $\mathbf{Q}$ be composed of column vectors $\{\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_n\}$. $\mathbf{Q}\mathbf{Q}^* = \sum_{i=1}^{n} (\mathbf{q}_i \mathbf{q}_i^*)$ is an orthogonal projection matrix.

Unitary matrices

Unitary Matrices

Definition

A matrix is unitary if $\mathbf{Q}^* = \mathbf{Q}^{-1}$,i.e. if $\mathbf{Q}^*\mathbf{Q} = \mathbf{Q}\mathbf{Q}^* = I$

- In the real case, we say the matrix is *orthogonal*. Its column vectors are *orthonormal*.
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Question: What is the geometric meaning of multiplication by a unitary matrix?
Answer: It preserves angles and Euclidean length. In the real case, multiplication by an orthogonal matrix $\mathbf{Q}$ is a rotation (if $\det(\mathbf{Q}) = 1$) or reflection (if $\det(\mathbf{Q}) = -1$).

Vector Norms

Definition of Norms

- Norm captures "size" of vector or "distance" between vectors
- There are many different measures for "sizes" but a norm must satisfy some requirements:

Definition

A norm is a function $||\cdot||$: $\mathbb{C}^m \to \mathbb{R}$ that assigns a real-valued length to each vector. It must satisfy the following conditions:

1. $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0$ only if $\mathbf{x} = 0$
2. $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
3. $\|\alpha\mathbf{x}\| = |\alpha|\,\|\mathbf{x}\|$.

- An example is Euclidean length (i.e.$||\mathbf{x}|| = \sqrt{\sum_{i=1}^{m} |x_i|^2}$)

p-norms

- Euclidean length is a special case of p-norms, defined as

$$\|\mathbf{x}\|_p = (\sum_{i=1}^{m} |x_i|^p)^{\frac{1}{p}}$$

for $1 \leq p \leq \infty$

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- 1-norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^{m} |x_i|$
- ∞-norm: $\|\mathbf{x}\|_\infty$. What is its value? Answer: $\|\mathbf{x}\|\,|_\infty = \max_{1 \leq i \leq m} |x_i|$
- Why we require $p \geq 1$? What happens if $0 \leq p < 1$?

Weighted p-norms

- A generalization of p-norm is weighted p-norm, which assigns different weights (priorities) to different components.
  - It is anisotropic instead of isotropic
- Algebraically, $\|\mathbf{x}\|_{\mathbf{W}} = \|\mathbf{W}\mathbf{x}\|$, where $\mathbf{W}$ is diagonal matrix with $i-$th diagonal entry $w_i \neq 0$ being weight for ith component
- In other words,

$$\|\mathbf{x}\|_{\mathbf{W}} = (\sum_{i=1}^{m} |w_i x_i|^p)^{\frac{1}{p}}$$

- What happens if we allow $w_i = 0$?

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- What happens if we allow $w_i = 0$?
- Can we further generalize it to allow $\mathbf{W}$ being arbitrary matrix?
- No. But we can allow $\mathbf{W}$ to be arbitrary nonsingular matrix.